

第七周 (2)

簡述學習目標

- (IBM Skillbuild)認識 clustering 演算法，了解如何利用機器學習的方式，幫助分析師在資安場景中過濾並篩選出異常行為，了解演算法在不同場景的適用性，並觀察資料特徵如何影響 AI 的分群判斷
- (hugging face)使用程式實作分群演算法，從 hugging face 取得資料或生成合成資料，透過分群結果學習辨識資料不平衡與雜訊，提升資料品質與分析效率
- (llama factory)進行基礎 SFT fine-tuning，讓模型模仿特定資安 Schema，透過實驗觀察資料分佈不均如何影響模型行為，理解高品質資料對微調結果的重要性

TASK 1 (Achieving): 學習 Clustering 原理與資安場景思考

作業內容

1. 完成 IBM 學習路徑: [Machine Learning for Developers](#)
2. 請寫下你的學習心得，並特別針對 clustering 演算法描述你在學習完後的理解
3. 結合你在課程中學到的機器學習概念，說明你認為哪種clustering方法可以如何運用在哪種資安場景中？
4. 說明可以用機器學習解決什麼樣的資安問題

截圖內容

- IBM 課程完成證明截圖(此次連結為學習路徑，故請同學截圖實作部分的學習過程做為學習佐證，謝謝)

TASK 2 (Exceeding): Clustering 實作

作業內容

1. 根據你在 task1 中想到的資安應用場景與合適的clustering，去 hugging face 或 kaggle 尋找合適的資料集
2. 撰寫clustering程式碼，實作將大量的資安資料進行歸類
實作參考連結：
 - Scikitlearn 官方: <https://scikit-learn.org/stable/modules/clustering.html#k-means>
 - <https://machinelearningmastery.com/document-clustering-with-llm-embeddings-in-scikit-learn/>
3. 觀察分群後的結果，紀錄你對結果的發現

截圖內容

- Clustering 實作過程

- 分群結果截圖

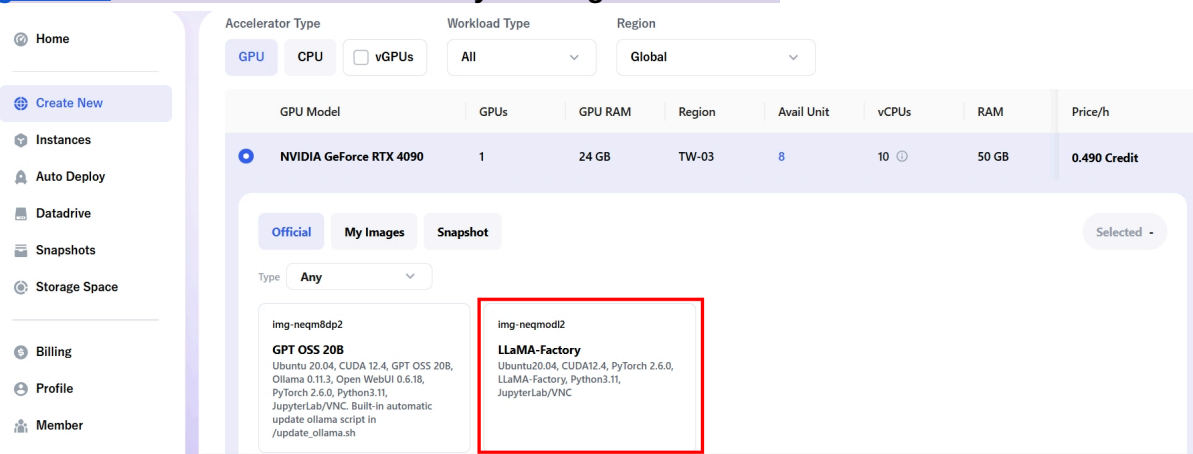
TASK 3 (Outstanding) : Llama-Factory 小微調實作

作業內容

1. 將轉換好的資料集餵入 Llama-Factory 進行 SFT 微調
2. 針對 task 2 分群結果出的結果分別進行提問，觀察在輸出格式的完整性以及內容的邏輯正確性上有何差異
3. 說明資料量的影響，如何決定模型在面對不同資安問題時，維持輸出格式穩定與避免產生幻覺的能力

llama-factory github 連結: <https://github.com/hiyouga/LlamaFactory>

glows.ai 內會開啟裝有 llama-factory 的 image 供同學使用



截圖內容

- 轉換後符合 Llama-Factory 格式的 json 資料片段截圖
- Llama-Factory 訓練參數與 loss 曲線截圖
- 對比實驗結果截圖: 對比資料充足 vs. 資料稀缺類別的 Schema 輸出差異

Week 7 (2)

Learning Objectives

- [\(IBM SkillBuild\)](#) Understand clustering algorithms and how machine learning can assist security analysts in filtering and identifying anomalous behaviors in cybersecurity scenarios. Students will explore how different clustering methods apply to different contexts and observe how data characteristics influence AI clustering decisions.
- [\(Hugging Face\)](#) Implement clustering algorithms through programming. Students will obtain datasets from Hugging Face or generate synthetic data, and analyze clustering results to understand issues such as data imbalance and noise, improving their awareness of data quality and analysis efficiency.
- [\(Llama-Factory\)](#) Perform basic SFT (Supervised Fine-Tuning) to train a model to mimic a specific cybersecurity schema. Through experiments, students will observe how uneven data distribution affects model behavior and understand the importance of high-quality datasets for fine-tuning results.

TASK 1 (Achieving): Understanding Clustering and Cybersecurity Applications

Assignment Content

1. Complete the IBM learning path : [Machine Learning for Developers](#)
2. Write a short reflection describing your learning experience. In particular, explain your understanding of clustering algorithms after completing the course.
3. Based on the machine learning concepts covered in class, explain: Which clustering method you think is suitable for a specific cybersecurity scenario.
4. How clustering can be applied in cybersecurity analysis. What kinds of cybersecurity problems could potentially be addressed using machine learning.

Screenshots to Include

- IBM course completion certificate(This link provides the learning path. Please include screenshots of your progress or completed activities as evidence of your learning process. Thank you.)

TASK 2 (Exceeding): Clustering Implementation

Assignment Content

1. Based on the cybersecurity application scenario you proposed in Task 1, select an appropriate clustering method. Search for a suitable dataset from Hugging Face or Kaggle related to your scenario.
2. Implement a clustering algorithm in Python to group or categorize large amounts of cybersecurity data.

Reference Resources :

- Scikit-learn documentation :
<https://scikit-learn.org/stable/modules/clustering.html#k-means>
 - Example tutorial :
<https://machinelearningmastery.com/document-clustering-with-llm-embedding-s-in-scikit-learn/>
3. Observe and analyze the clustering results, and record your findings.

Screenshots to Include

- clustering implementation process
- clustering results

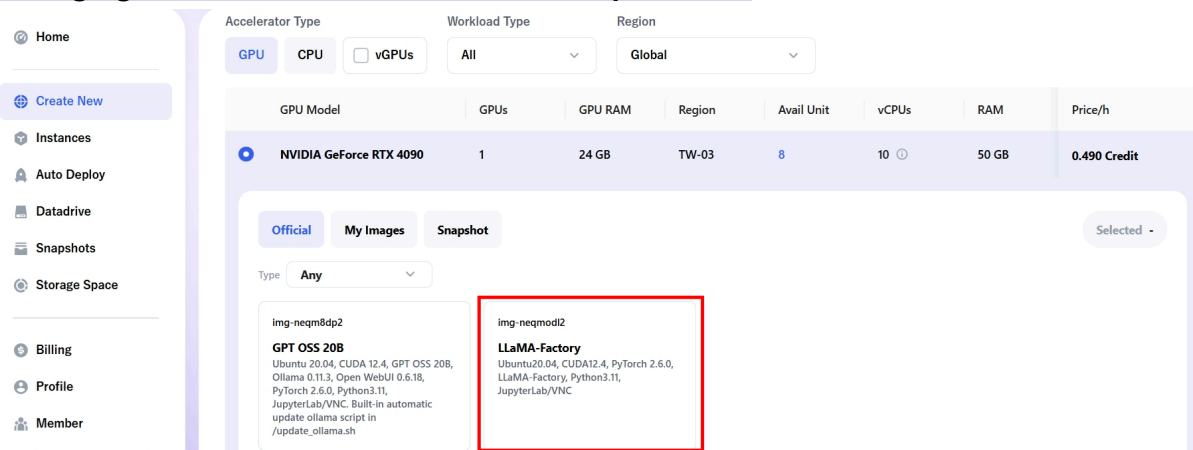
TASK 3 (Outstanding): Llama-Factory Fine-Tuning Experiment

Assignment Content

1. Use the converted dataset to perform Supervised Fine-Tuning with Llama-Factory.
2. Based on the clustering results obtained in Task 2, create several test queries and evaluate the model outputs. Observe and analyze: differences in output format completeness, differences in logical correctness of responses
3. Explain how dataset size and distribution influence the model's ability to maintain consistent output formats and reduce hallucinations when responding to cybersecurity-related queries

Llama-Factory GitHub repository : <https://github.com/hiyoga/LlamaFactory>

A preconfigured container image with Llama-Factory installed will be provided through glows.ai for students to run their experiments



Screenshots to Include

- converted dataset in Llama-Factory JSON format
- training parameters and loss curve
- comparison between data-rich vs data-scarce categories and their impact on schema output quality